



METHOD ARTICLE

REVISED Direct oligonucleotide sequencing with nanopores

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Abstract

Third-generation DNA sequencing has enabled sequencing of long, unamplified DNA fragments with minimal steps. Direct sequencing of ssDNA or RNA gives valuable insights like base-level modifications, phosphoramidite synthesis yield estimates and strand quality analysis, without the need to add the complementary strand. Direct sequencing of single-stranded nucleic acid species is challenging as they are non-compatible to the double-stranded sequencing adapters used by manufacturers. The MinION platform from Oxford Nanopore Technologies performs sequencing by passing single-strands of DNA through a layer of biological nanopore sensors; although sequencing is performed on single-strands, the recommended template by the manufacturer is double-stranded. We have identified that the MinION platform can perform sequencing of short, single-strand oligonucleotides directly without amplification or second-strand synthesis by performing a single annealing step before library preparation. Short 5' phosphorylated oligos when annealed to an adapter sequence can be directly sequenced in the 5' to 3' direction via nanopores. Adapter sequences were designed to bind to the 5' end of the oligos and to leave a 3' adenosine overhang after binding to their target. The 3' adenosine overhang of the adapter and the terminal phosphate makes the 5' end of the oligo analogous to an end-prepared dsDNA, rendering it compatible with ligation-based library preparation for sequencing. An oligo-pool containing 42,000, 120 nt orthogonal sequences was phosphorylated and sequenced using this method and ~90% of these sequences were recovered with high accuracy using BLAST. In the nanopore raw data, we have identified that empty signals can be wrongly identified as a valid read by the MinION platform and sometimes multiple signals containing several strands can be fused into a single raw sequence file due to segmentation faults in the software. This direct oligonucleotide sequencing method enables novel applications in DNA data storage systems where short oligonucleotides are the primary information carriers.

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Keywords

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REVISED Amendments from Version 1

Information on AMXpINS3 and AMXpEINS3 sequences has been added and discussed upon. The fast5 raw data images have been annotated and discussed. Discussion on sequencing of 3' unblocked oligos has been edited and expanded. The occurrence of reads with artifacts has been discussed and explained. The blastn results are visualized in 2D histograms for each of the samples. The omission of the bead clean-up step and its impact on the flowcell performance has been discussed.

Any further responses from the reviewers can be found at the end of the article

Plain language summary

Traditional DNA sequencing methods need the target DNA to be double-stranded. The MinION platform from Oxford Nanopore Technologies performs sequencing by analyzing the voltage fluctuations while passing a single strand of DNA/RNA through a nanopore. In the current method, single-stranded DNA strands were modified to mimic double-stranded DNA by modifying their ends. The 5' end of the DNA strands are converted into a double strand to facilitate the attachment of the sequencing adapters. To our surprise, sequencing with nanopores did not require the strands to be double-stranded. The modifications of the DNA ends are sufficient to facilitate sequencing of single-stranded nucleic acids. A pool of short, single-stranded DNA molecules was successfully sequenced using this method. The raw data from the nanopore sequencer has shown that the internal software was unable to properly splice the sequence data in some instances. The method to sequence short, single-stranded DNA (oligonucleotides) directly without making them double stranded could be applied in DNA-data storage applications.

Introduction

DNA sequencing has become a staple tool in biology, has become affordable and more accessible to small labs and individual researchers in the past decade. Oxford Nanopore Technologies (ONT), with its biological nanopore-based sequencing technology, has opened the market wide open

by releasing a \$1000 sequencing platform – The MinION¹. The MinION platform has the capability to sequence both amplified and non-amplified double-stranded DNA (dsDNA)² and direct RNA in the 3' to 5' direction using Poly-A tail capture³. Direct sequencing of short, single-stranded oligonucleotides has been regarded as a challenge due to nanopore chemistry, pore design and basecalling⁴. However, attempts have been made to overcome these challenges by producing long templates consisting of concatemeric repeats of short target sequences⁵. Direct sequencing of short, single-strand nucleic acid species without a polymerase or ligation step has not been evident in previous research⁶. In this article we propose a method to perform direct sequencing of short single strand oligonucleotides that can be leveraged by different applications such as DNA-based data storage systems and direct RNA sequencing.

In DNA-based data storage systems, oligonucleotides are information carriers and rapid sequencing of oligos is needed to extract the encoded data⁷. Several encoding and compression techniques are used to design oligonucleotides for DNA data storage purposes to increase the data capacity and to deal with amplification and sequencing issues. The possibility to sequence oligonucleotides directly without performing a PCR step or performing a ligation step enables users to design the oligos to have only one priming region on the 5' end, freeing up nucleotide space used for the reverse priming region. This increases the encoding space available to the users and opens possibilities for new encoding schema and DNA data storage architectures. In this work, we propose a method that is incredibly fast when compared to PCR-based sequencing strategies⁸ with hands-on time as low as five minutes and sequencing time of just 20 minutes for pre-phosphorylated oligos like INS3 and EINS3 shown in [Figure 1](#). We have identified that the Oxford Nanopores MinION platform is capable of sequencing single-strand templates directly without the need for a complementary strand or to have a spacer strand to increase the strand length.

In this work, we solve this challenge of direct oligonucleotide sequencing by performing a simple annealing step before the

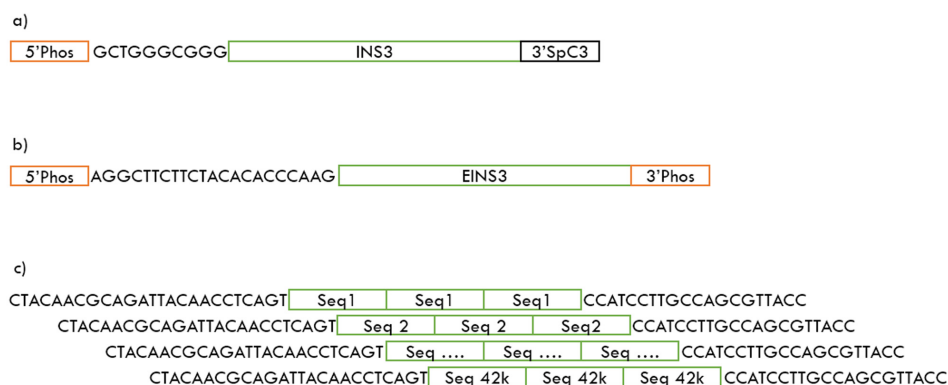


Figure 1. Sequencing templates **a)** INS3 oligo (175 nt) containing a 5' phosphate and 3' C3 spacer **b)** EINS3 oligo (200 nt) containing a 5' phosphate and 3' terminal phosphate **c)** 3x6 oligo-pool containing 42,000 unique oligonucleotides of 120 nt length, each of these oligo contain a 3x repetitive region of a 25-nt orthogonal sequence. All sequences in the 3x6 oligo-pool contain 5' and 3' priming regions.

library preparation step. The setup starts with a phosphorylation step using T4 Polynucleotide Kinase (PNK) that adds a 5' phosphate to the target oligos. The phosphorylated oligos are annealed to an adapter sequence that binds to their 5' end. The adapter sequences are designed to have a melting temperature of ~65°C to the 5' end of the target oligos; and when annealed to their targets, the adapter strands have an adenosine overhang at their 3' end, as detailed in Figure 2. In the annealed state with the adapter sequence, the 5' end of the oligos are analogous to an end-prepared dsDNA and are compatible with the AMX sequencing adapters from the ligation sequencing kit (LSK-109) offered by ONT⁹. The helicase-bound sequencing adapter (AMX) has a thymine (T) overhang on the 3' end of its top-strand and a recessed 5' phosphate end on its bottom-strand as shown in Figure 2 (b). The sequences used to implement the method and related protocols are discussed in the next section.

Methods

Materials and oligonucleotides

Three sets of oligonucleotides were used in this study: INS3, EINS3 and 3xr6. The INS3 and EINS3 sequences were procured from IDT from their Ultramer manufacturing line, both sequences were 5' phosphorylated at synthesis. INS3 has a 3' C3 spacer terminator and EINS3 has a 3' phosphate that were also added during manufacturing. An oligo pool (3xr6) containing 42,000 unique, 120-nt sequences was procured from Twist Biosciences. The annealing adapters INS3 RC, EINS3 RC and ArcFP were procured from IDT and these oligos did not contain any modifications. All the oligos procured from IDT were normalized to 100 uM concentration in TE buffer. The 3xr6 oligo pool was normalized to 10 ng/ul concentration in TE buffer. Please refer to Supplementary Table 1 (see *Extended data*) for sequence data and reagent details¹⁰. A step-by-step version of the protocol is available at protocols.io: <https://dx.doi.org/10.17504/protocols.io.bt84nryw>.

Normalization and Qubit analysis

The 3xr6 oligo pool was normalized to 0.25 uM in TE buffer by calculating the average molecular weight of the individual oligos and the final concentration was verified using Qubit ssDNA Assay Kit (Cat# Q10212) on a Qubit 4 instrument. The INS3 and EINS3 oligos were diluted to 0.5 uM concentration using

TE buffer. The INS3 RC, EINS3 RC and the ArcFP oligos were diluted to their 1 uM working concentrations.

Phosphorylation and sequencing of 3xr6

The 3xr6 oligo pool was phosphorylated using T4 PNK kinase from NEB. The phosphorylation was carried out at 2 picomole template concentration for 30 minutes at 37°C for 30 minutes followed by a heat inactivation step at 65°C for 20 minutes. The phosphorylated oligo pool was reconcentrated using Monarch spin columns from NEB using the manufacturers oligo cleanup protocol and eluted in TE buffer. The elute from the spin wash was three-way split for a triplicate sequencing run. 1 picomole of the annealing adaptor (ArcFP) was added to each of the triplicate and the temperature was raised to 94°C for 2 minutes for strand denaturation and the mixture was slowly cooled to room temperature for target binding. The sequencing adaptor - AMX from ONT and the Blunt/TA mastermix from NEB were added to each of the triplicates to carry out the ligation reaction between the sequencing adaptor and the oligo pool. The final mixtures were loaded on to three MinION flow cells and the sequencing was performed for 4 hours.

Library preparation and sequencing of INS3 and EINS3

A triplicate of 0.25 picomoles of INS3 mixed with 0.5 picomoles of INS3 RC in nuclease free water were denatured at 94°C for 2 minutes and slowly cooled to room temperature. The same setup is followed for EINS3 and EINS3 RC with final reaction volumes of 3 ul. The sequencing adaptor (AMX) and Blunt/TA mastermix were added to the reaction tubes to facilitate the ligation reaction. Three flow cells were used for the INS3 triplicate sequencing run and the flowcells are washed as per the manufacturers protocol and the three flow cells were again used for the EINS3 triplicate run. Each of the triplicate sequencing runs were carried out for 20 minutes.

Sequencing run and basecalling

The sequencing runs were carried out on r9.4 MinION flow cells from ONT at default settings on MinKNOW for the ligation sequencing kit. The manufacturer recommends the DNA to be cleaned before loading on to the flowcell, but in this study the oligos were directly loaded alongside the ligase without a wash step. The sequencing runs were carried out for

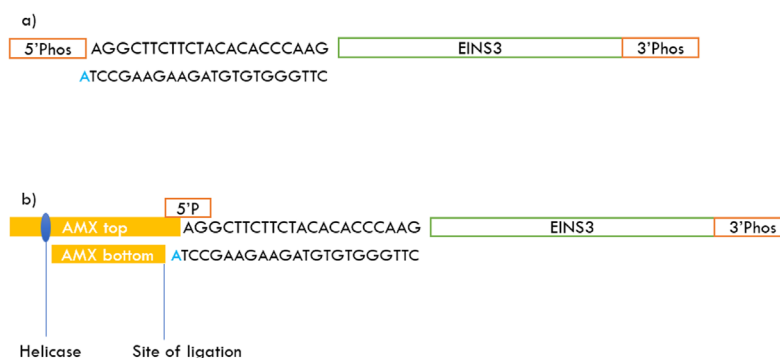


Figure 2. Annealing and ligation steps **a)** Annealing of the adapter to the 5' end of the EINS3 strand and leaving an adenosine overhang **b)** Ligation of the AMX sequencing adapter to the adapter + EINS3 strand.

4 hours in the case of the 3xr6 oligo pool and for 20 minutes each in the case of INS3 and EINS3. The fast5 raw signal files were basecalled on a laptop with dedicated Nvidia GPU (RTX 2060) using GPU-Guppy (ver. 3.5.2) in the high accuracy mode¹¹. The generated fastq files were binned into pass or fail folders based on their q-scores. Only the reads that have passed the q-score threshold were analyzed.

BLASTN analysis and FAST5 visualization

BLAST-N program from NCBI was run locally with different modes. The default mode was run for identifying longer matches, whereas the short regions were identified using the 'blastn-short' flag. 99-percentile matches were calculated based on their high identity and accuracy scores to the reference sequences. The blast analysis was performed on a 6-core Intel CPU with 12 threads. The fast5 raw data was visualized with HDFView (Ver 3.1.0) to identify and understand the morphology of the single-stranded DNA sequencing.

Results

Sequencing of INS3

The flowcells 1, 2 and 3 yielded 23248, 15917 and 21052 reads, respectively, after 20 minutes of sequencing and the reads were basecalled. A total of 289, 483 and 255 reads from flowcells 1 to 3 passed the default q-score filter. The sequencing and BLASTN results with e-value filter are shown in Table 1. AMXpINS3 sequence is generated by concatenating the last 10 bases at 3' end of AMX adaptor with the first 10 bases at the 5' end of INS3 oligo. AMXpINS3 reflects the resultant sequence after the sequencing adaptor (AMX) has been ligated to INS3.

Sequencing of EINS3

The flowcells 1, 2 and 3 yielded 23174, 30238, and 30051 reads, respectively, after 20 minutes of sequencing and the

generated reads were basecalled. A total of 6489, 5041 and 3826 reads from flowcells 1 to 3 passed the default q-score filter. The sequencing and BLASTN results with e-value filter are shown in Table 2. AMXpEINS3 sequence is generated by concatenating the last 10 bases at 3' end of AMX adaptor with the first 10 bases at 5' end of EINS3 oligo. AMXpEINS3 reflects the resultant sequence after the sequencing adaptor (AMX) has been ligated to EINS3.

Sequencing of 3xr6

The flowcells 1, 2 and 3 yielded 60299, 48432 and 42434 reads, respectively, after four hours of sequencing followed up by basecalling. A total of 28869, 13299 and 10268 reads from flowcells 1 to 3 passed the default q-score filter. The sequencing and BLASTN results with e-value filter are shown in Table 3. AMXpFP is generated by concatenating the last 10 bases at the 3' end of the AMX adaptor with the first 10 bases of the universal priming region at the 5' end of 3xr6 oligos. AMXpFP reflects the resultant sequence after the sequencing adaptor (AMX) has been ligated to the oligos in 3xr6 pool. FPpRP sequence is generated by concatenating the 10 bases at the 3' end of the oligos in the 3xr6 pool with the 10 bases at the 5' end of the oligos in 3xr6 pool. FPpRP sequences can arise when the oligos within the 3xr6 pool ligate to one another during the library preparation.

Data analysis

BLASTN analysis of INS3, EINS3 and 3xr6 are shown in Figure 4, Figure 5 and Figure 6, respectively, which show the total number of reads that pass the quality threshold and the number of significant matches that are found. For INS3 and EINS3, the input query is their full sequence, and the total number of significant matches are plotted. The significant matches are searched with default BLASTN parameters. High-quality matches are also plotted with a 99-percentile

Table 1. INS3 sequencing results.

	Total reads	Total reads passing q-score filter	Filter pass %	All significant matches	Significant matches to INS3 (e value < 1e-65)	Significant matches to AMXpINS3 (ACGTATTGCTGCTGGGCGGG) "blastn-short"
Flowcell 1	23248	289	1.24	281	4	693
Flowcell 2	15917	483	3.03	414	11	820
Flowcell 3	21052	255	1.21	228	4	635

Table 2. EINS3 sequencing results.

	Total reads	Total reads passing q-score filter	Filter pass %	All significant matches	Significant matches to eins3 (e value < 1e -75)	Significant matches to AMXpEINS3 (ACGTATTGCTAGGCTTCTTC) "blastn-short"
Flowcell 1	23174	6489	28.00	5922	143	10363
Flowcell 2	30238	5041	16.67	4767	53	10264
Flowcell 3	30051	3826	12.73	3699	34	8966

Table 3. 3xr6 sequencing results.

	Total reads	Total reads passing q-score filter	Filter pass %	At least 1 significant match vs 42k targets (e value < 1e -15)	At least 1 significant match vs 42k addresses (e value < 1e -02) "blastn-short"	Significant matches to AMXpFP (ACGTATTGCTCTACAACGCA) "blastn-short"	Potential fusion matches FPPRP (CAGCGTTACCTACAACGCA) "blastn-short"
<i>Flowcell 1</i>	60299	28869	47.877	25177	31436	88862	78
<i>Flowcell 2</i>	48432	13299	27.46	12146	19128	35789	39
<i>Flowcell 3</i>	42434	10268	24.2	10374	16678	30605	25
<i>Combined</i>	151165	52436	34.69	32028	37265	155256	142

label in the figures; these matches show very-high identity to the search query. Refer to the BLASTN output files available as underlying data (see *Data availability* section).

The data from 3xr6 sequencing run is analyzed with BLASTN to search for the full-length sequences along with the short 25-nt orthogonal sequence. The 25-nt orthogonal sequences used during the design of 3xr6 oligo-pool are taken directly from a published source¹². Each of the 42,000 (120-nt) oligos in the 3xr6 oligo pool contain a unique 25-nt orthogonal sequence that is repeated three times within the same strand (3x25-nt = 75-nt). All sequences in the 3xr6 oligo pool contain the same forward and reverse priming regions for PCR-compatibility. The 3xr6 oligo-pool was not amplified prior to sequencing in this study.

Discussion

The technique of modifying the 5' end of an oligo to make it compatible for nanopore sequencing has resulted in some interesting insights into the sequencing mechanism. The helicase-bound sequencing adapter (AMX) has a thymine (T) overhang on its top-strand and an oligo with a 5' phosphate and a short (10-nt) 5'-end double-strand region with an adenosine (A) overhang that can facilitate sequencing. The biological nanopore used by ONT's MinION system can process a single-stranded template without the impeding force provided by the complementary strand. Although the helicase modifications and the nanopore mechanism is proprietary, we believe that the voltage gradient was the driving force behind the strand translocation through the R9.4.1 nanopore flowcells. The helicase may or may not be functional but could be impeding the strand translocation and slowing it enough to perform a high-resolution scan through a k-mer.

We have also identified that 3' unblocked oligonucleotides like the oligos in the 3xr6 pool can be sequenced by the same method. The INS3 and EINS3 sequences are designed using

human insulin gene template¹³ and manufactured by phosphoramidite process. Both INS3 and EINS3 strands have a 5' phosphate and 3' blocker molecules added during their synthesis. The oligos in the 3xr6 oligo-pool are phosphorylated using T4 PNK before annealing to its adapter sequence. The 3' end of the oligos in 3xr6 are unprotected, unlike INS3 and EINS3, which can lead to concatenated products during the ligation step of library preparation.

We have skipped the bead-based clean-up step of the samples after the sequencing adaptor ligation, as we intended to load as much as the sample we could and to shorten the hands-on-time for sequencing. The omission of the clean-up step could be a reason for obtaining reads with just the AMX adaptors like seen in [Figure 3](#) as the bead clean-up removes small oligos from the sample. We believe that a modified clean-up step with high bead to sample ratio could select and purify short oligos while removing the non-ligated AMX adaptor. A clean-up step could also improve the number of reads obtained during sequencing as the cross contaminants from the library step like the ligase and free AMX adaptors might have lowered the pore occupancy of the actual reads and/or fouled the nanopores.

We have observed that several of the reads that are generated by the sequencer are empty without any viable signal data as visualized in [Figure 3](#) using HDFView (Ver 3.1.0)¹⁴, these reads contain several stall events and helicase dissociation events that are evident from the spike signals. The low scoring read that has been shown in [Figure 3](#) do not contain any target (INS3) sequences, but only the free AMX sequencing adaptors. A repeating pattern of initial broad spike denoting helicase association with the nanopore followed by a short signal representing the AMX adaptors sequence and a final short, high amplitude spike denoting the dissociation of the adaptor from the nanopore. The low scoring read in [Figure 3](#) shows 7 such events in its entire length and did not pass the quality filter of the basecalling.

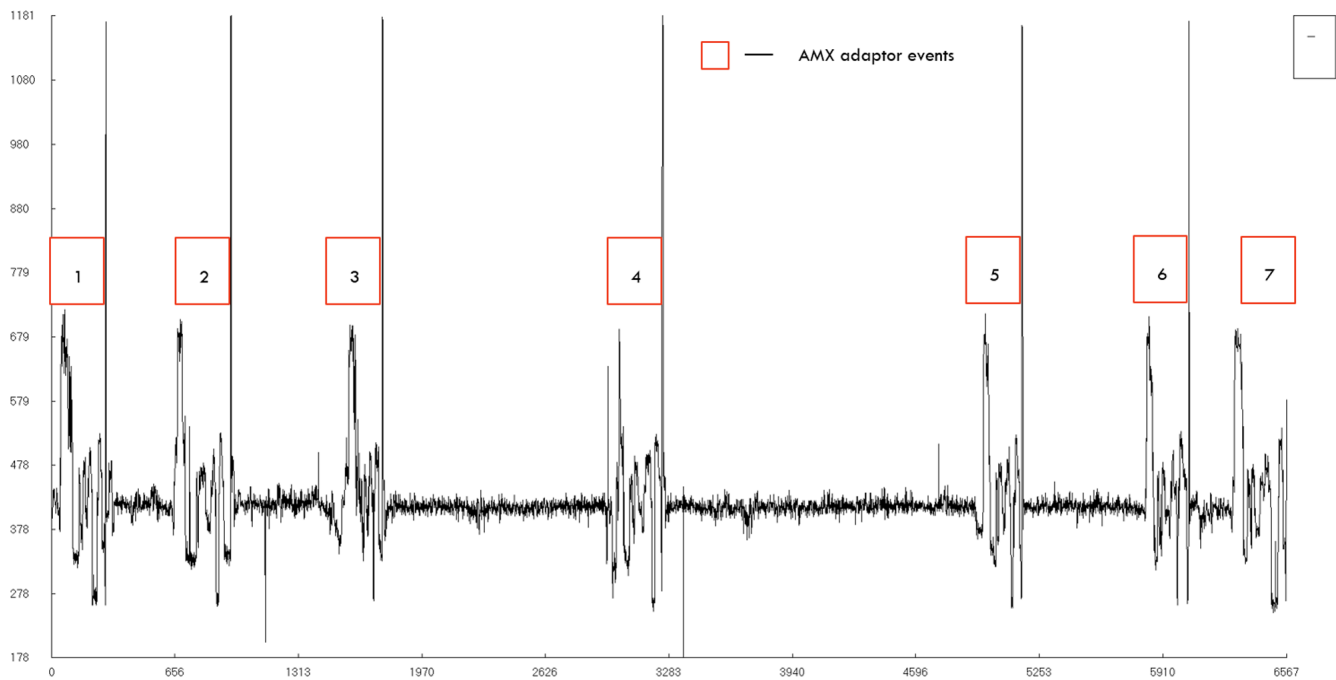


Figure 3. Visualization of a low-scoring INS3 (flowcell 1) read.

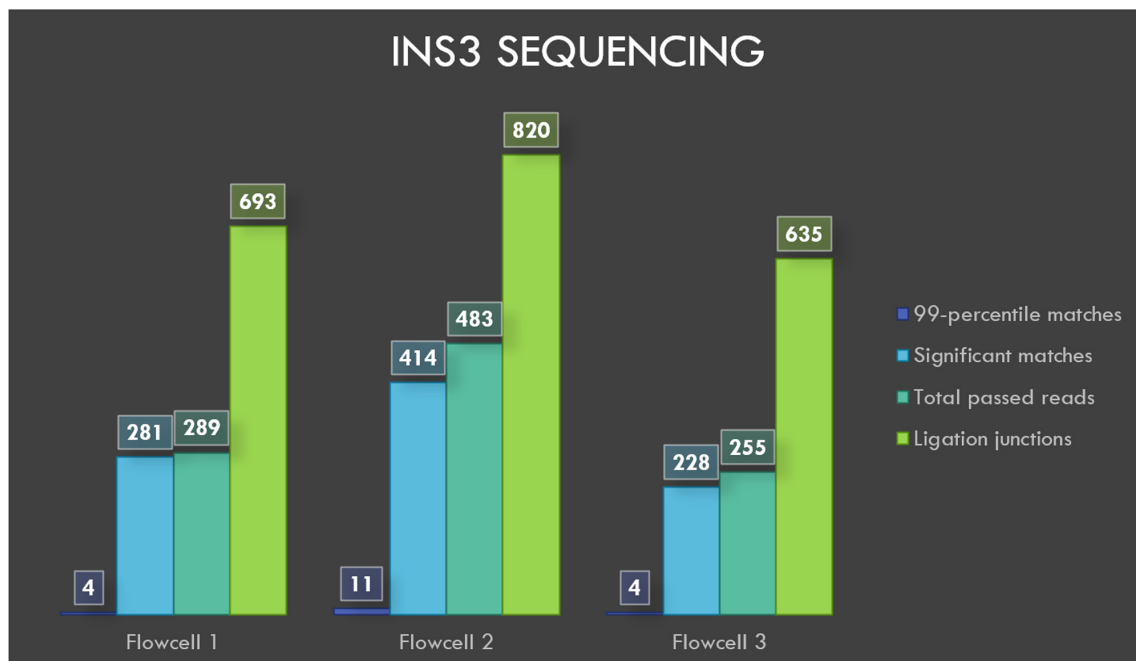


Figure 4. Visualization of INS3 sequencing results.

Reads like the one shown in [Figure 3](#) highlights the issues with the data processing on the MinION platform where non-reads could be classified as reads and reads like shown in

[Figure 7](#), which are not split properly for high accuracy base-calling. Reads that pass the Guppy quality check are basecalled and analyzed using BLASTN¹⁵.

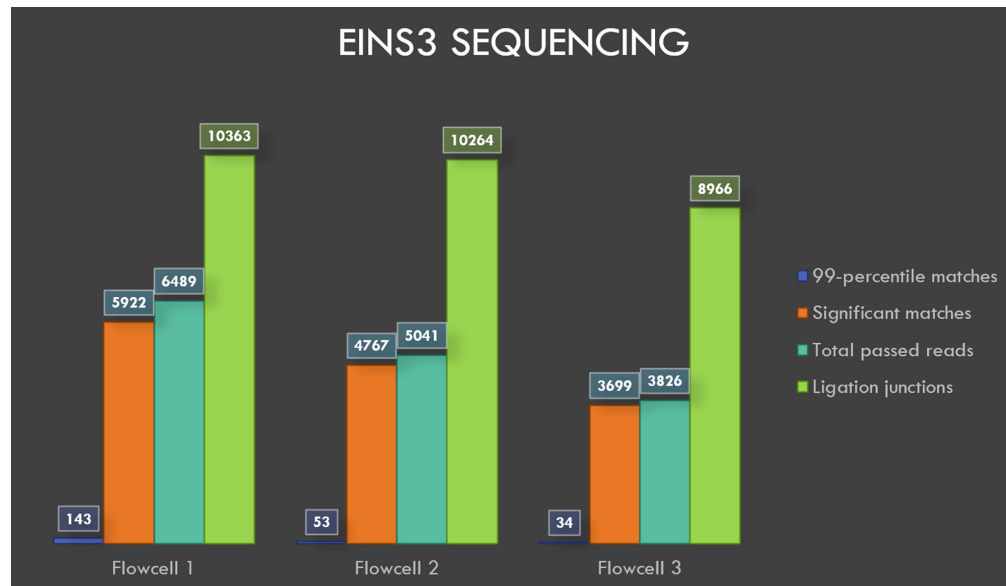


Figure 5. Visualization of EINS3 sequencing results.

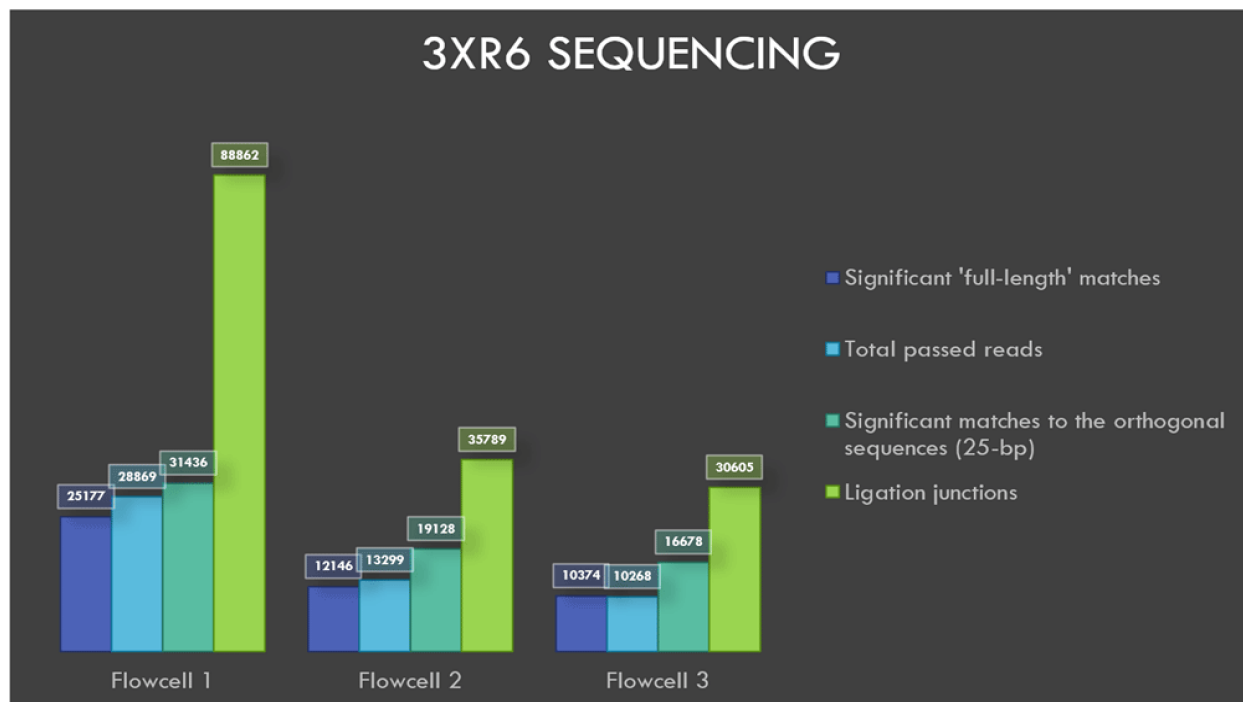


Figure 6. Visualization of 3xr6 sequencing results.

Ligation junctions and multi-strand reads

The ligation junctions where the AMX adapter binds to the INS3, EINS3 and 3xr6 oligos are searched using BLASTN with

‘blastn-short’ flag. The junction sequences AMXpINS3, AMXpINS3 and AMXpFP as shown in the [Table 1](#), [Table 2](#) and [Table 3](#), respectively, are plotted in [Figure 4–Figure 6](#). The

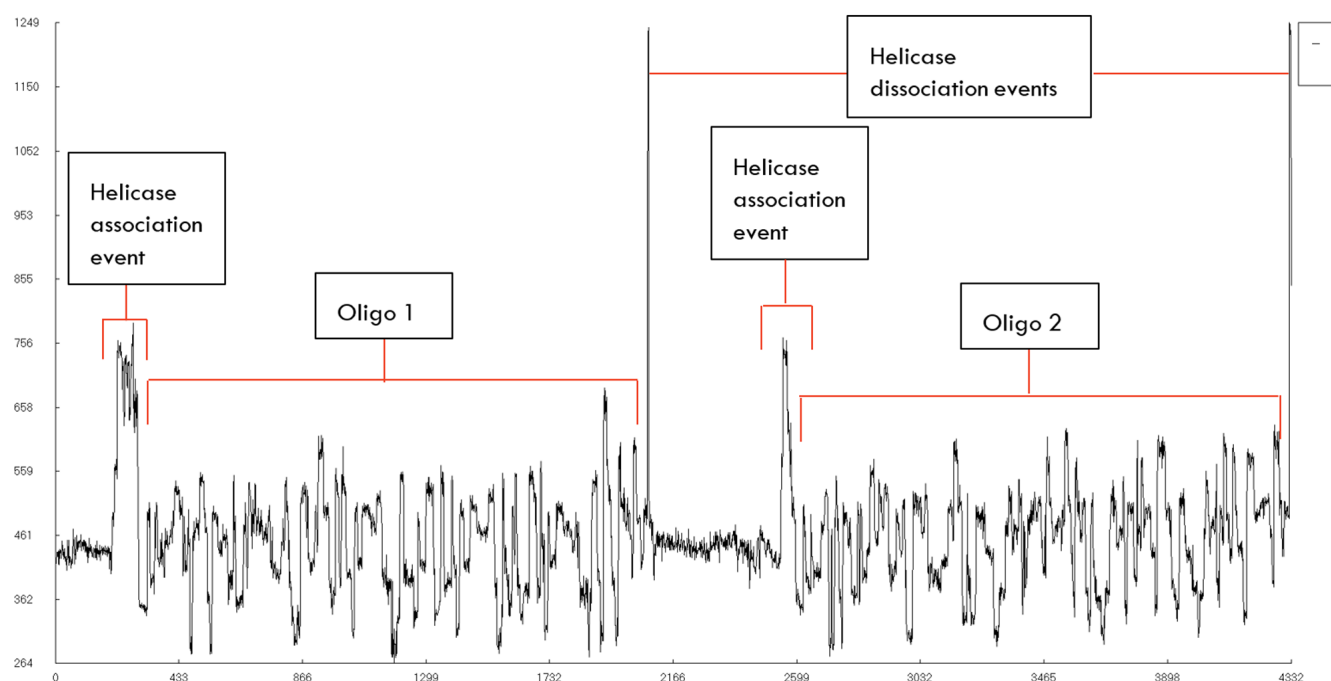


Figure 7. Visualization of a fusion read – [7a6c8a08-3d36-4fcf-a765-ef9cf3e28188] from 3xr6 sequencing (flowcell 1).

number of ligation junctions (AMXpINS3, AMXpINS3 and AMXpFP) are found in excess than the actual number of reads due to the read artifacts where each read may contain more than one strand. The ligation junctions are also found in significantly higher number than the full-length sequences due to their short length (20 nt), relative to the lengths of INS3, EINS3 and the oligos in the 3xr6 pool. We have identified several of these multi-strand reads, and visualization of such a read is provided in Figure 7.

Conclusion

We have proposed and implemented a method to sequence single-strand nucleic acid species without performing amplification, second-strand synthesis or a spacer ligation step with helicase-based biological nanopores offered by ONT. Oligonucleotides with a free 3'-OH can also be sequenced successfully with the described method. We have identified sequencing artifacts during our data analysis. Some empty signals can be earmarked as valid reads by the MinION data processing system and reads with signal information for more than one strand can also be observed as with the case of Figure 7, where two distinct signals for two reads can be found in its raw data. Reads like the ones shown in Figure 7 have low quality score due to the spike regions within them and are needed to be split at their helicase dissociation event to have high quality basecalling.

We believe that this sequencing approach might open new avenues for DNA-based information storage systems and lead to improvements in signal-level data analysis for nanopore sequencing.

Data availability

Underlying data

Imperial College London – Research Data Repository: Direct oligonucleotide sequencing with nanopores, <https://doi.org/10.14469/hpc/8139>¹⁰.

This project contains the following underlying data:

- Raw fast5 files from INS3 sequencing run in 1a.zip, 2b.zip and 3d.zip
- Raw fast5 files from EINS3 sequencing run in eins_1a.zip, eins_2b.zip and eins_3d.zip
- Raw fast5 files from 3xr6 sequencing run in 1a-42k(split).zip, 2b-42k(split).zip and 3d-42k(split).zip
- Basecalled fasta files and BLASTN analysis data from the sequencing runs in data.zip

Extended data

Imperial College London – Research Data Repository: Direct oligonucleotide sequencing with nanopores, <https://doi.org/10.14469/hpc/8139>¹⁰.

This project contains the following extended data:

- Supplementary_Table_1.tsv (Supplementary Table 1)

Data are available under the terms of the [Creative Commons Attribution 4.0 International license](#) (CC-BY 4.0).

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Fumihito Miura 

Department of Biochemistry, Kyushu University Graduate School of Medical Sciences, Fukuoka, Japan

The authors have adequately addressed my concerns. I am fully satisfied with the changes and the authors' responses.

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Genomics, biotechnology, epigenetics

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Reviewer Report 27 August 2021

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Jeff Nivala 

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Karen Zhang

Molecular Information Systems Lab, University of Washington, Seattle, WA, USA

The authors have addressed our concerns.

Competing Interests: JN is a consultant to ONT.

Reviewer Expertise: nanopore technologist

We confirm that we have read this submission and believe that we have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Version 1

Reviewer Report 27 July 2021

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This manuscript describes an approach to sequence 5' phosphorylated ssDNA using commercial nanopore sequencing (ONT). The trick is to anneal an oligo onto the 5' end of the ssDNA such that there is an A-overhang suitable for TA-based ligation of the dsDNA ONT sequencing adapter onto the ssDNA-oligo complex. Some comments and questions:

- The success of this approach is somewhat foreseeable considering that ONT-based 2D and 1D-squared style sequencing demonstrate the helicase motor can process ssDNA through the pore.
- Overall, the results have been presented clearly for the most part, and interpretation of the results seems sound.
- We agree with the authors, it does simplify the workflow for sequencing of large oligo pools, eliminating the need for PCR.
- It was also interesting that the authors omitted the purification step following adapter ligation, this also simplifies readout workflow considerably.
- Possible related to this is the relatively low read counts, and low number of reads passing filter. It would be great if the authors could discuss this a bit more. Is it possible this is due

to committing the cleanup step?

- For those not used to looking at raw nanopore traces, it would help to annotate them with the phenomena that the authors are trying to highlight.

Is the rationale for developing the new method (or application) clearly explained?

Yes

Is the description of the method technically sound?

Yes

Are sufficient details provided to allow replication of the method development and its use by others?

Yes

If any results are presented, are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions about the method and its performance adequately supported by the findings presented in the article?

Yes

Competing Interests: Jeff Nivala is a consultant to Oxford Nanopore Technologies.

Reviewer Expertise: nanopore technologist

We confirm that we have read this submission and believe that we have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however we have significant reservations, as outlined above.

Reviewer Report 30 June 2021

<https://doi.org/10.21956/openreseurope.14648.r27140>

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Fumihito Miura

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The authors presented a method to prepare sequencing libraries for Oxford Nanopore's

sequencers from short oligonucleotides by partially forming double-stranded DNA for adapter tagging with dsDNA ligation.

The ideas and results seem good. However, as a whole, their data presentations and explanations are poor.

Major

1. In Tables 1 and 2, why the numbers of significant matches to INS3 and EINS3 are lower than the AMXpINS3 and AMXpEINS3, respectively? In my understanding, the AMXpINS3 and AMXpEINS3 contain INS3 and EINS3, respectively, and BLASTN uses local alignment. Therefore, BLASTN should detect any partially matched sequences between the reads and INS3 or EINS3. The authors should provide explanations.
2. What did the authors want to explain with Figures 3 and 7? Only raw data is presented in these figures. The authors should give explanations for stalls and helicase dissociations in Figure 3. Also, they should explain what signals indicate the multi-strand reads in Figure 7.
3. In Figures 4, 5, and 6, the authors used bar charts in 3D. However, while they provide the numbers at the top of the bars, the 3D chart is difficult to compare the values. Therefore, they should use 2D charts in these figures.
4. In the Discussion section, the authors mentioned that "We have also identified that the oligonucleotides can even be 3' unblocked if they do not contain a terminal thymine (T)". What did they mean by this sentence? I could not find any relevant description and data in the Result section for this description.
5. In the last part of the Discussion section, the authors mentioned that "reads containing several strands per read are also found, which could be because of segmentation errors." What did they mean with the "segmentation errors." The authors meant the base caller crash? Or simply program error?

Minor

1. What is the "AMX sequencing adaptor" in the Introduction section? While I can find the explanation in the Discussion section, the authors should explain it the place first appeared.
2. There is no explanation for AMXpINS3 and AMXpEINS3.

Is the rationale for developing the new method (or application) clearly explained?

Yes

Is the description of the method technically sound?

Yes

Are sufficient details provided to allow replication of the method development and its use by others?

Yes

If any results are presented, are all the source data underlying the results available to

ensure full reproducibility?

Yes

Are the conclusions about the method and its performance adequately supported by the findings presented in the article?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Genomics, biotechnology, epigenetics

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 19 Jul 2021

Sachin Chalapati

We agree to the points that the reviewer has made and will make the appropriate revisions to the relevant sections, nevertheless we would like to clarify on them in this comment for reference.

1. AMXpINS3 and AMXpEINS3 are short sequences containing the 3' end of the sequencing adaptor (AMX) and the 5' end of INS3 and EINS3 respectively. They are short sequences that do not contain the entirety of INS3 and EINS3 and were found in excess compared to the significant matches of INS3 and EINS3.
2. We have found the helicase dissociation events and multi-strand reads in raw fast5 files, but the data is inconsistent on why they are occurring and the root-cause of these issues. It was observed that in some instances, there can be reads without any valid sequencing data, or instances where there are several sequences within a single fast5 read. We believe that we can rescue the multi-strand reads by splitting them into individual strands and basecalling them. Relevant information would be added to the discussion section.
3. We agree with the reviewer's suggestion and make changes to the charts.
4. The INS3 and EINS3 strands have a blocked 3' ends, as to eliminate any spurious ligations taking place during the adaptor ligation step of the library prep. The 3xr6 oligo pool do not have any 3' end modifications but were sequenced successfully with the same protocol. We were suggesting that the 3' end can be unblocked and still be sequenced. The line will be corrected appropriately.
5. We believe the multi-strand reads occur due to segmentation issues on the instrument (MinION), where the sequencer is failing to split a read correctly. In general, the nanopore reads would only contain one sequence per read.
6. Information on the sequencing adaptor and other short sequences would be added to the relevant sections.

Competing Interests: No competing interests were disclosed.